**SPL-1 Project Proposal Form, 2026**
**Institute of Information Technology (IIT)**
**University of Dhaka**

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## MiniGit: A Simplified Version Control System

**Project Description:**

The proposed project is an offline, command-line based **Version Control System** inspired by Git, focused on a single, linear project history. The system will allow users to track file changes, stage selected files, create commits, view commit history, compare different versions, restore previous versions, and merge changes.

**Features to be implemented:**

- **Repository Initialization (init):** Creates a new local repository and sets up the necessary internal structure for storing project data, file information, and version history.
- **File Tracking and Staging (add):** Allows users to select new or modified files and add their current changes to the staging area, preparing them to be included in the next commit.
- **Commit (commit):** Creates a new saved version of the project from the staged changes, along with a commit message and information needed to maintain the project's version history.
- **Status (status):** Shows the current state of project files, including untracked files, modified files, and files that have already been staged for committing**.**
- **Commit History (log):** Displays previously created commits with their commit information and relationships, allowing users to examine how the project has changed over time**.**
- **Version Comparison (diff):** Compares file contents between versions or the current working state and shows which lines have been added, removed, or modified.
- **Hashing:** Uses hashing to uniquely identify stored file contents and determine whether the contents of a file have changed between different versions**.**
- **Branching (branch):** Allows users to create and manage separate lines of development from existing commits, so different changes can be developed independently.
- **Branch Switching (checkout**): Allows users to switch between branches and updates the working project files to match the selected branch's version**.**
- **Merging (merge):** Combines changes from two different branches into a single branch and detects conflicting changes when both branches modify the same parts of a file.
- **Error Handling:** Provides appropriate handling and messages for invalid commands, missing files, invalid commit or branch references, and other common operations that cannot be completed**.**

**Languages or Tools to be used:**

- C++ programming language
- Git and GitHub for version control
- Linux / WSL command line
- Hashing library/ implementation

**Supervisor's Name:** Mridha Md. Nafis Fuad

**Signature of the supervisor:** _____

**Date:** 23/09/2026

**Before the Midterm Presentation:**
I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.
**Signature of the supervisor:** _____
**Date:** _____

**Midterm Presentation Feedback:**

**Before the Final Presentation:**
I confirm that the project is complete and I am forwarding it for final presentation.
**Signature of the supervisor:** _____
**Date:** _____